25900335 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSECFS200 - TCP/IP AND NETWORK SECURITY (SPECIALIZATION : NETWORK SECURITY)

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1. Recognize the role of port numbers in end-to-end communication. [K] / [CO2]
2. Define multiplexing in transport layer context. [K] / [CO1]
3. Define Computer Networks. [K] / [CO1]
4. Differentiate TCP and UDP. [U] / [CO1]
5. Name the three rules of Bell-La Padula model [K] / [CO2]
6. Name the different types of firewall [K] / [CO2]
7. Write the two ways to protect password file [A] / [CO3]
8. List the advantage of symmetric encryption. [K] / [CO3]
9. Differentiate between authentication and authorization. [U] / [CO3]
10. Define digital signature. [K] / [CO3]
11. List the components of the IPSec protocol suite. [K] / [CO4]

12. State one difference between ESP and AH. [K] / [CO4]

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **6** questions

Each question carries **5** marks

13. Explain IP address. [U] / [CO2]

14. Classify email protocols and state their uses. [U] / [CO2]

15. Explain different types of Passive Attacks through example [U] / [CO2]

16. Inspect the charecteristics of firewall [An] / [CO4]

17. Explain how tickets are used in Kerberos authentication. [U] / [CO3]

18. State the role of Certification Authority (CA) and Registration Authority (RA). [K] / [CO3]

19. Classify the features of SSL and TLS based on security strength and protocol versions. [A] / [CO4]

20. Compare SSH and TLS in terms of security features. [U] / [CO4]

[5x6 = 30]

Part C

Essay Type Questions

Answer any **2** questions

Each question carries **10** marks

21. Illustrate Chinese wall model in detail with a figure [A] / [CO2]

22. Explain the importance of X.509 certificates in secure communication. [U] / [CO3]

23. Produce an integrated email security solution using both PGP and S/MIME. [A] / [CO4]

[10x2 = 20]